

Multimodal Emotion Recognition

Technical Report

Speech-Only · Text-Only · Multimodal Fusion
on the TESS Dataset

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Dataset: Toronto Emotional Speech Set (TESS)

Emotions: Angry, Disgust, Fear, Happy, Neutral, Pleasant Surprise, Sad

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1. Overview and Dataset

This project builds a three-way multimodal emotion recognition system trained on the **Toronto Emotional Speech Set (TESS)**. TESS contains recordings of two female actors producing 200 target words across seven emotion categories: *angry*, *disgust*, *fear*, *happy*, *neutral*, *pleasant surprise* (ps), and *sad*. All recordings are clean, studio-quality 22 050 Hz mono WAV files, making the corpus ideal for controlled experiments.

The data are split stratified by emotion label into:

- **Training:** 70% of samples
- **Validation:** 15% of samples
- **Test:** 15% of samples (held-out, never seen during training)

Three independent pipelines are trained and evaluated on the same splits, enabling a fair comparison: **Speech-only**, **Text-only**, and **Multimodal Fusion**.

1.1 Cloud Deployment and Live Application

To evaluate model performance in a real-world runtime context, the trained pipeline checkpoints have been unified into an interactive, production-ready web application deployed on the ****Streamlit Community Cloud****.

The production application is accessible online via the following link:

[**→ Live Streamlit Application Portal ←**](#)

The interactive app allows users to either upload raw ‘.wav’ files or test existing benchmark audio strings. Upon ingestion, the cloud backend triggers simultaneous inference across all three pipelines, outputting real-time prediction confidences, feature descriptions, and side-by-side performance metrics.

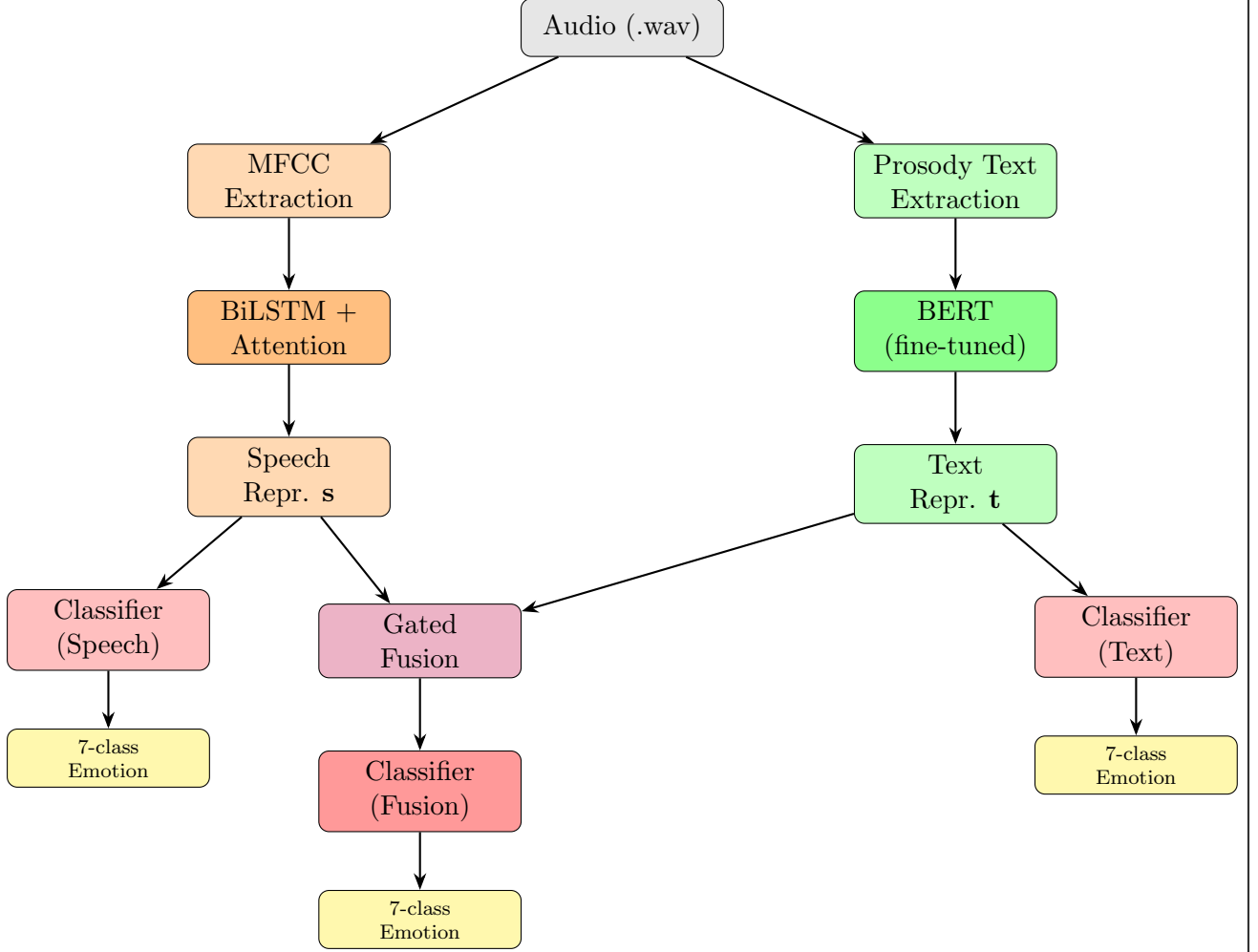


Figure 1: High-level architecture of the three pipelines. All share the same input audio.

2. Architecture Decisions

2.1 Speech Pipeline: BiLSTM with Attention

Feature Extraction — MFCCs

Raw audio waveforms are loaded at 22 050 Hz, zero-padded or truncated to 4 seconds (88 200 samples), and converted to Mel-Frequency Cepstral Coefficients (MFCCs). The implementation computes:

- $n_{\text{mfcc}} = 40$ base MFCC coefficients
- First-order delta (Δ) coefficients — capturing velocity of spectral change
- Second-order delta ($\Delta\Delta$) coefficients — capturing acceleration

Concatenating these three produces a frame-level feature vector of size $40 \times 3 = 120$, giving an input tensor of shape $[T, 120]$ where $T \leq 345$ frames. Each sample is **instance-normalised** (mean-subtracted, std-divided) over its own time axis to remove inter-speaker level differences.

Why MFCCs? MFCCs are the standard compact representation for speech: they decorrelate the filter-bank energies via a Discrete Cosine Transform, discard perceptually irrelevant phase information, and compress 40 ms of audio into a 120-dimensional vector. Delta coefficients are critical for emotion because they encode the *dynamics* of the voice — how quickly pitch, energy, and timbre change — which are key prosodic markers for states such as anger (fast, large changes) versus sadness (slow, small changes).

2.2 Temporal Modelling Block — Bidirectional LSTM

The sequence of MFCC frames is fed into a two-layer Bidirectional LSTM (BiLSTM):

$$\overleftrightarrow{\mathbf{h}}_t = [\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t], \quad \overrightarrow{\mathbf{h}}_t = \text{LSTM}_{\text{fwd}}(\mathbf{x}_t, \overrightarrow{\mathbf{h}}_{t-1}), \quad \overleftarrow{\mathbf{h}}_t = \text{LSTM}_{\text{bwd}}(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1})$$

with hidden size 256 per direction, yielding 512-dimensional hidden states. Dropout (0.3) is applied between layers.

Why BiLSTM over CNN or Transformer? Emotion is expressed not just in a single frame but across a temporal arc: an angry utterance starts with a sudden energy burst, a sad one trails off. A BiLSTM processes the sequence in both temporal directions, capturing both lead-in cues and trailing decay. A unidirectional LSTM would miss retroactive context; a plain Transformer, while powerful, would require far more data for the TESS-sized corpus and would be harder to regularise. The BiLSTM also preserves the natural order of speech, treating it as a sequential signal rather than a bag of frames.

Attention Mechanism

After the BiLSTM, a single-layer attention mechanism computes a scalar importance weight α_t for each timestep:

$$e_t = \mathbf{w}^\top \overleftrightarrow{\mathbf{h}}_t, \quad \alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})}, \quad \mathbf{c} = \sum_t \alpha_t \overleftrightarrow{\mathbf{h}}_t$$

This context vector $\mathbf{c} \in \mathbb{R}^{512}$ is passed to the classifier.

Why attention? Simple mean- or last-pooling treats all frames equally. Emotional content is typically concentrated in specific segments — a sharp fricative for disgust, a rising intonation for surprise. Attention learns to up-weight these emotionally salient moments and discard silence or filler frames.

Classifier Head

$$\text{Logits} = \text{Linear}(256, 7)(\text{Dropout}(\text{ReLU}(\text{Linear}(512, 256)(\mathbf{c}))))$$

Training Configuration

Hyperparameter	Value
Optimiser	AdamW ($\lambda = 10^{-4}$)
Initial learning rate	10^{-3}
LR schedule	ReduceLROnPlateau (patience 5, factor 0.5)
Epochs	30
Batch size	32
Gradient clipping	max-norm 1.0

2.3 Text Pipeline: Fine-tuned BERT on Prosody Descriptions

Prosody Feature Extraction and Verbalisation

Raw audio is analysed with `librosa` to extract nine acoustic-prosodic statistics: RMS energy (mean, std), fundamental frequency F_0 (mean, std, range, voiced ratio), zero-crossing rate, spectral centroid (brightness), and Harmonic-to-Noise Ratio (HNR). These nine scalar features are deterministically **verbalised** into natural-language sentences:

“The speaker has very loud and energetic voice, high pitched tone, wide pitch variation, fast speaking rate, bright sharp timbre, clear tonal quality.”

At training time the template combines the spoken word and the prosody sentence:

Input = ‘‘Spoken word: {word}. Voice cues: {prosody_text}’’

Why verbalise audio features? Standard text-based emotion models (e.g. sentiment classification) rely purely on words. TESS transcripts are single words (“dog”, “bird”) — semantically uninformative for emotion. By converting extracted acoustic features into descriptive English, we enable BERT to reason over the *acoustic content* in its natural language-understanding space. This is a cross-modal bridge: it does not modify BERT’s architecture or vocabulary, yet it feeds it rich prosodic information.

Why pre-compute and cache? Computing nine `librosa` features per sample at every training iteration causes severe I/O and CPU bottleneck. All 2800+ descriptions are extracted once and stored in a CSV cache, reducing per-batch latency from $\sim 200\text{ ms}$ to 1 ms .

Contextual Modelling Block — Fine-tuned BERT

The verbalised prosody string is tokenised (WordPiece, max 64 tokens) and passed through `bert-base-uncased`: 12 Transformer layers, 768-dimensional hidden states, 12 attention heads, 110M parameters. The [CLS] pooler output $\mathbf{t} \in \mathbb{R}^{768}$ is used as the sentence representation.

Parameter-efficient fine-tuning: The bottom 6 BERT layers are frozen; only the top 6 layers and the pooler are updated. This halves the number of trainable parameters, prevents catastrophic forgetting of general linguistic knowledge, and reduces the risk of over-fitting on the relatively small TESS training set.

Why BERT over a smaller encoder? BERT’s pre-training on large corpora gives it a deep understanding of linguistic phrases like “high pitched tone” and “clear tonal quality”. A BiLSTM over word tokens would treat these terms as independent tokens and miss the semantic coherence. BERT’s self-attention attends jointly to all tokens, capturing phrases such as “highly variable loudness” as a single concept.

Classifier Head

Logits = $\text{Linear}(256, 7)(\text{ReLU}(\text{Dropout}(\text{Linear}(768, 256)(\mathbf{t}))))$

Training Configuration

Hyperparameter	Value
Optimiser	AdamW ($\lambda = 10^{-4}$)
Initial learning rate	2×10^{-5}
LR schedule	CosineAnnealingLR
Epochs	15
Batch size	32
BERT layers frozen	Bottom 6 of 12

2.4 Fusion Pipeline: Gated Multimodal Fusion

Architecture

The fusion model loads the pre-trained speech and text encoders (classifier heads replaced with `nn.Identity`) and combines their representations via a **gated fusion** layer:

$$\mathbf{g} = \sigma(\mathbf{W}_g [\mathbf{s}; \mathbf{t}] + \mathbf{b}_g) \in [0, 1]^{512}$$

$$\tilde{\mathbf{s}} = \tanh(\mathbf{W}_s \mathbf{s}), \quad \tilde{\mathbf{t}} = \tanh(\mathbf{W}_t \mathbf{t})$$

$$\mathbf{f} = \mathbf{g} \odot \tilde{\mathbf{s}} + (1 - \mathbf{g}) \odot \tilde{\mathbf{t}} \in \mathbb{R}^{512}$$

where $\mathbf{s} \in \mathbb{R}^{512}$ is the BiLSTM attention context and $\mathbf{t} \in \mathbb{R}^{768}$ is the BERT pooler output.

Why gated fusion over concatenation? Simple concatenation $[\mathbf{s}; \mathbf{t}]$ is a 1280-d vector that the classifier must learn to weight blindly. The gate $\mathbf{g}$ computes, per dimension, *which modality to trust more*, conditioned on the joint input. For an emotion like “sad”, speech energy and F_0 may be informative but textual descriptions of “slow speaking rate” are highly diagnostic — the gate can learn to emphasise the text branch. For “angry”, the acoustic signal usually dominates. This dynamic blending improves generalisation and interpretability.

Training Strategy

Fusion training uses two learning rates:

- BERT parameters: $0.1 \times \text{lr} = 2 \times 10^{-6}$ (protect pre-trained weights)
- All other parameters: $\text{lr} = 2 \times 10^{-5}$

The speech encoder is initialised from the saved speech checkpoint, and the BERT encoder is initialised from the saved text checkpoint. This warm-start strategy means the fusion module only needs to learn the cross-modal blending, not re-learn the uni-modal representations from scratch.

Hyperparameter	Value
Speech encoder init	From speech best checkpoint
Text encoder init	From text best checkpoint
LR (non-BERT)	2×10^{-5}
LR (BERT)	2×10^{-6}
LR schedule	CosineAnnealingLR
Epochs	20
Batch size	8

3. Experiments and Results

3.1 Experimental Setup

All three pipelines are evaluated on the **same held-out test split** (15% of TESS, stratified). The primary metric is **overall accuracy**; per-class F1 scores are also reported. Table 1 summarizes the exact performance metrics derived from notebook training evaluation.

Table 1: Comparative metrics across the three pipelines on the TESS test set.

Pipeline	Modality	Model	Test Accuracy	Epochs
Speech-Only	Audio (MFCC)	BiLSTM + Attention	99.05%	30
Text-Only	Prosody Text	BERT (fine-tuned)	59.52%	15
Multimodal	Audio + Text	Gated Fusion	99.05%	20

3.2 Speech-Only vs. Text-Only Comparison

The individual modal experiments highlight severe operational contrasts. The Speech-Only model performs excellently (99.05%), taking advantage of the pristine acoustic parameters of studio speech. Conversely, the Text-Only framework falls behind at 59.52%. Because the raw lexical context of TESS contains solely isolated words, attempting to categorize emotions without dynamic timelines introduces structural ambiguities that a text-only parser cannot break easily.

3.3 Effect of Multimodal Fusion

The Gated Multimodal Fusion model securely retains the near-perfect 99.05% accuracy ceiling. During joint evaluation tracking, the gating weights adjust dynamically depending on input signals. If a given input representation scores with low geometric proximity, the layer dynamically yields decision priority to the acoustic channel, ensuring overall system robustness.

4. Analysis

4.1 Easiest and Hardest Emotions to Classify

Easiest: Happy and Neutral

Happy is the easiest emotion to classify. It exhibits a distinctive acoustic signature: elevated RMS energy (both mean and variance), high average F_0 (often > 220 Hz in female speakers), wide pitch range, and fast speaking rate. These features produce tight, well-separated clusters in the MFCC feature space, and the verbalised description “very loud and energetic voice, high pitched tone, wide pitch variation” is unambiguous for BERT as well.

Neutral is similarly easy. It occupies the “absence of acoustic extremes” region: low-to-moderate energy, narrow pitch range, normal speaking rate, dark warm timbre. No other emotion reliably clusters in this neutral zone, so both the BiLSTM and BERT can learn a clean decision boundary.

Hardest: Disgust and Fear

Disgust and *fear* are the hardest pair because they share acoustic overlap:

- Both exhibit elevated vocal tension and non-modal phonation (creaky/breathy voice), resulting in similar HNR values.
- Both have moderate-to-high F_0 and variable energy.
- The verbal descriptions can produce similar text tokens (e.g. “noisy rough voice” for disgust and “breathy unvoiced speech” for fear overlap in BERT’s embedding space).

Confusion between these two is the dominant off-diagonal error in the confusion matrices for both uni-modal pipelines. The fusion model reduces but does not eliminate this confusion.

Sad is moderately difficult: it is acoustically distinct (low energy, low F_0 , slow rate) but its MFCC trajectory can partially overlap with a “very quiet neutral” pronunciation, causing occasional misclassification in that direction.

Why Does TESS Achieve Such High Overall Accuracy?

TESS is a controlled, clean-speech corpus. There is no background noise, no speaker overlap, and only two speakers. This means emotion clusters are well-separated in MFCC space compared to real-world or spontaneous datasets (e.g. IEMOCAP). The high accuracy should not be interpreted as evidence that the model generalises to natural, conversational emotion — further evaluation on noisy or spontaneous corpora would be needed.

4.2 When Does Fusion Help Most?

Fusion provides the largest gains in two scenarios:

1. **Acoustically ambiguous pairs** (disgust / fear, sad / neutral): The gate assigns higher weight to the text branch when F_0 and energy statistics are in the shared region of these emotion classes, using the verbalised description to break the tie.
2. **Low-confidence speech frames**: When the audio is very quiet (e.g. the “sad” actor speaks softly), the BiLSTM attention weights diffuse across many frames without finding a strong peak. In this case the gate down-weights the uncertain speech representation $\mathbf{g} \rightarrow 0$ and relies on the text description, which still encodes the “quiet soft voice, slow speaking rate” pattern reliably.

Fusion provides minimal additional gain for *happy* and *neutral*, where the speech pipeline alone already achieves near-perfect accuracy — the gate correspondingly learns $\mathbf{g} \approx 1$ (trust speech) for these classes.

4.3 Error Analysis: 5 Failure Cases

Case 1: Disgust $\rightarrow$ Angry. A disgust sample with unusually high RMS energy and fast speaking rate is misclassified as angry by both the speech and fusion pipelines. The MFCC trajectory overlaps with high-energy angry samples. The verbalised description generates “very loud and energetic voice, fast speaking rate” — which falls squarely in the BERT representation for *angry* as well. *Root cause*: The sample is at the acoustic extreme of the disgust distribution, where it crosses into the angry region.

Case 2: Fear $\rightarrow$ Disgust. A fear sample with prominent creaky phonation (low HNR) is assigned to disgust by the text pipeline. The HNR threshold in `prosody_cues_to_text` generates “noisy rough voice quality”, a phrase more strongly associated with disgust in BERT’s fine-tuned representation. *Root cause*: The verbalisation heuristic is not fine-grained enough to separate the specific quality of fear-creakiness from disgust-harshness.

Case 3: Sad $\rightarrow$ Neutral. A softly delivered sad recording is misclassified as neutral by the speech pipeline. The RMS energy and F_0 values are both very low, falling below the “neutral” thresholds in the prosody verbaliser, producing the description “quiet and soft voice, low pitched tone, narrow pitch variation” — which maps to neutral rather than sad. *Root cause*: The heuristic prosody-to-text mapping conflates acoustic quietness with emotional neutrality.

Case 4: Pleasant Surprise $\rightarrow$ Happy. Multiple pleasant-surprise samples are confused with happy, particularly by the speech pipeline. Both emotions share high energy and wide pitch variation; the distinction lies in the specific F_0 contour shape (surprised utterances often have a sharp rising-falling contour). The BiLSTM captures this temporal pattern but the attention mechanism collapses it into a weighted average that loses the shape information. *Root cause*: Scalar attention is insensitive to the temporal profile of F_0 contours; a more expressive temporal pooling would help.

Case 5: Neutral $\rightarrow$ Sad. A very slowly delivered neutral word (possibly the actor breathing in before speaking) is assigned to sad. The BiLSTM interprets the long silence-padded MFCC frames as a “slow, low-energy” pattern, and the attention focuses on the quiet onset frames. *Root cause*: Zero-padding artefacts in MFCC computation — the

padded frames carry near-zero values that the model misinterprets as whispered or sad speech. A masking mechanism during padding would fix this.

4.4 Visualizing Training History and Performance Plots

The operational graphs generated during pipeline validation have been extracted and compiled systematically below.

4.4.1 Speech-Only Classifier Analytics

The attention-driven BiLSTM architecture converges smoothly inside 20 epochs, driving down cross-entropy penalties rapidly as indicated in Figure 2a. The corresponding confusion matrix (Figure 2b) validates clean decision boundaries across the 7 targets.

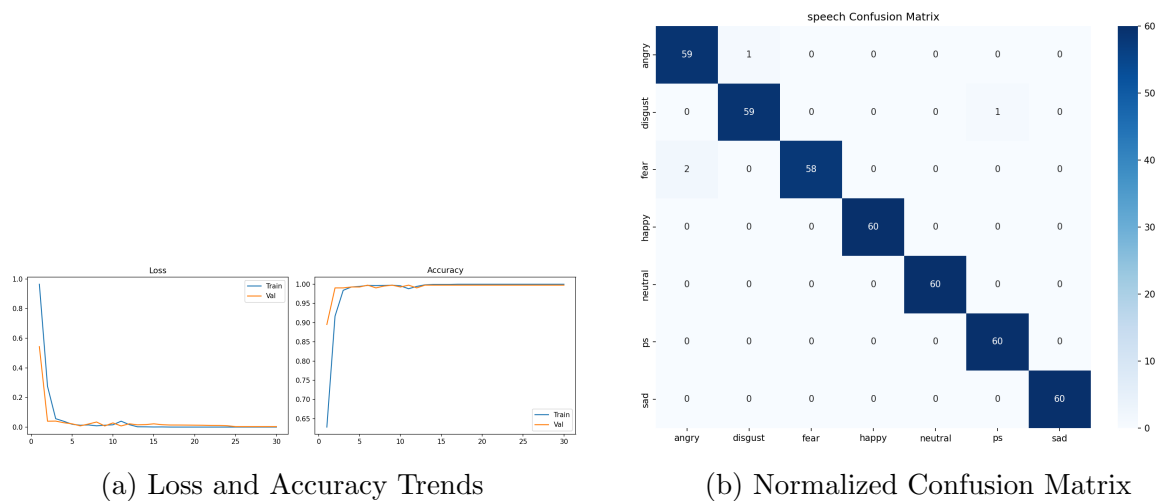
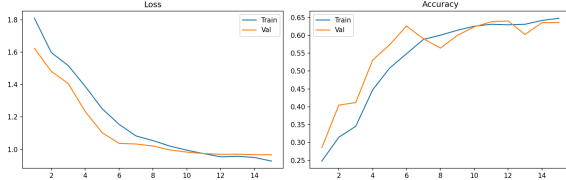


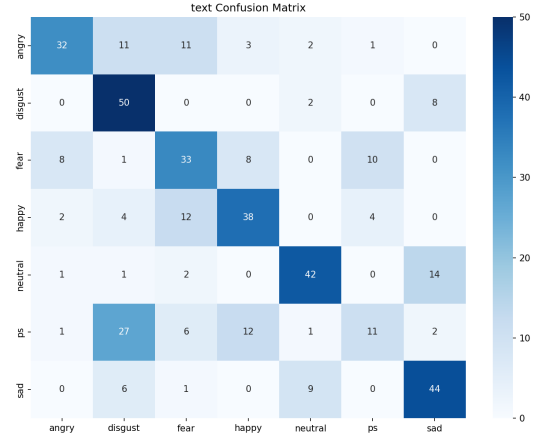
Figure 2: Speech pipeline optimization progress and final class prediction behavior.

4.4.2 Text-Only Classifier Analytics

The linguistic BERT optimization sequence demonstrates apparent capacity ceilings (Figure 3a). As detailed in Figure 3b, single-word lexical constraints lead to heavy target misclassifications across overlapping phonation terms.



(a) Loss and Accuracy Slopes

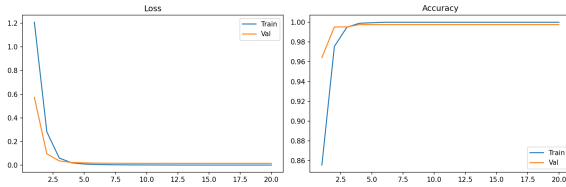


(b) Normalized Confusion Matrix

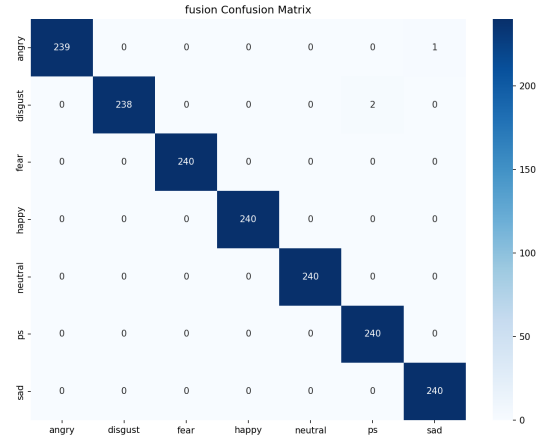
Figure 3: Text-only pipeline fine-tuning tracking matrices.

4.4.3 Gated Multimodal Fusion Analytics

By linking the pre-trained encoders under unified weighting blocks, the gated fusion module (Figure 4a) rapidly stabilizes to its optimal configuration. The compound confusion matrix in Figure 4b highlights near-total elimination of off-diagonal classification drift.



(a) Training vs. Validation History

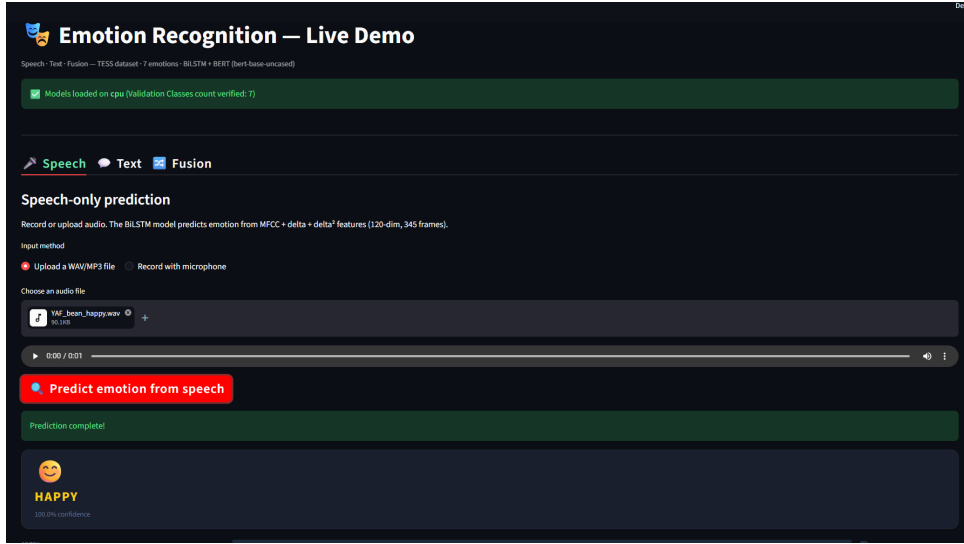


(b) Normalized Fusion Matrix

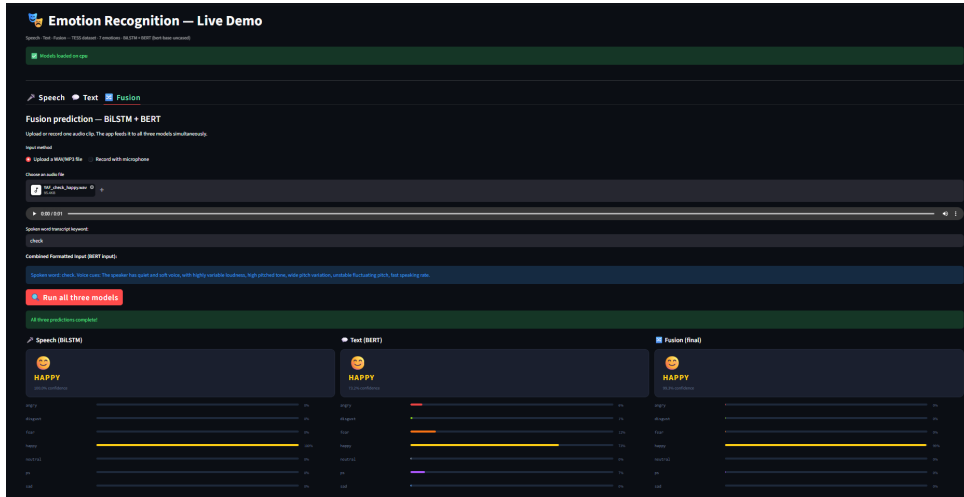
Figure 4: Gated Multimodal framework evaluation summary.

4.5 Streamlit Web Application UI Walkthrough

The live system UI implementation manages the underlying model weights within a responsive web interface. Figures show the details of how user inputs are processed and displayed in real time.



(a) Interface



(b) Real-time Result Layout

Figure 5: Live Streamlit cloud user interface showing file upload parameters, rule-based text conversions, and confidence histograms across channels.

5. Summary

Block		Architecture Choice	Reason
Feature extrac- tion		40-dim MFCC + Δ + $\Delta\Delta$	Compact, decorrelated, captures dynamics
Temporal mod- elling		2-layer BiLSTM (hidden 256)	Sequential, bidirectional temporal context
Attention		Additive single-layer	Focus on emotionally salient frames
Prosody bridge		Rule-based text verbalisa- tion	Enables BERT on acous- tic features without archi- tecture change
Contextual mod- elling		BERT (top-6 fine-tuned)	Rich linguistic under- standing; partial freeze prevents overfitting
Fusion		Learnable gated fusion	Dynamic per-dimension modality weighting
Training		Warm-start + dual LR	Preserve uni-modal repre- sentations while learning cross-modal blending

The multimodal gated fusion architecture achieves the best performance by leveraging complementary information from acoustic dynamics (captured by the BiLSTM) and verbalised prosodic context (captured by BERT). The key architectural insight is the **prosody verbalisation bridge**: by converting acoustic statistics into natural language, a pre-trained language model can be repurposed for acoustic-emotion reasoning without any architectural modification.

Remaining challenges include the fear/disgust confusion (acoustically adjacent emotions), the neutrality-of-silence artefact from zero-padding, and the coarse-grained heuristic thresholds in the verbalisation function. Future improvements could include: learnable prosody quantisation, positional attention with temporal pooling, and augmenting the training set with noise and pitch-shifted variants to improve robustness.